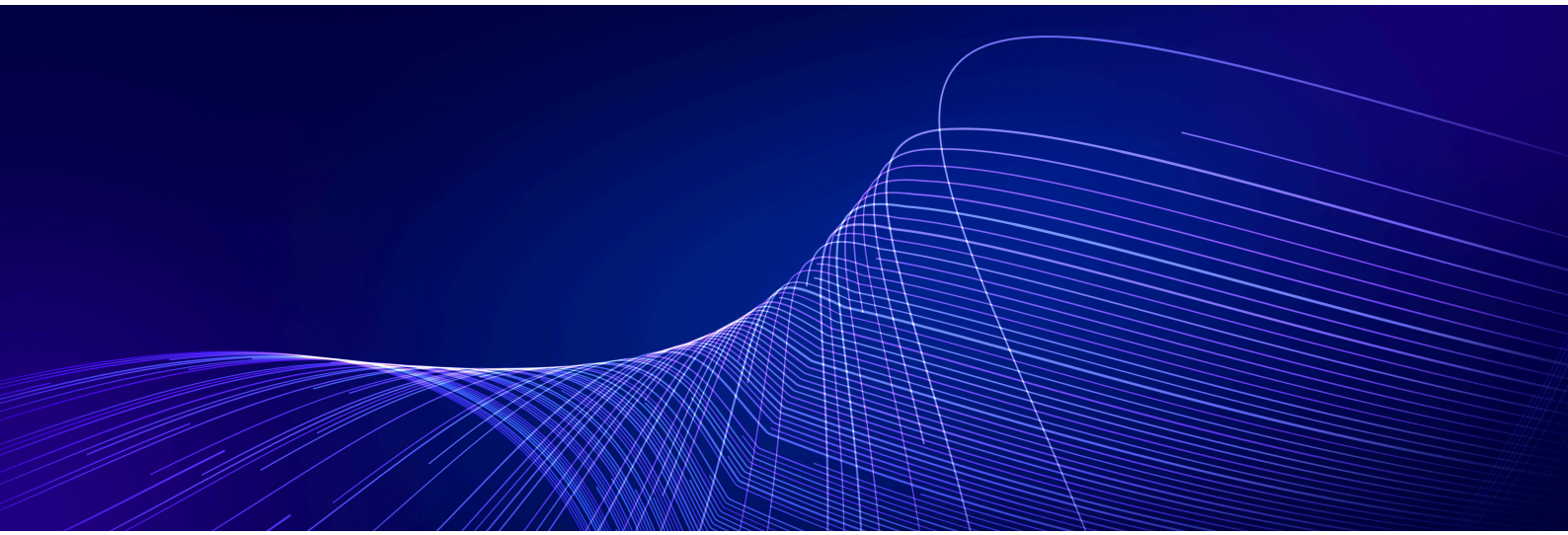

Data Science & Machine Learning

Course Outline

Phase 1: Foundations of Data Science & Python Programming

Duration: Weeks 1–2

Timeline	Topic	Key Focus Areas
Days 1–2	**Introduction to Data Science**	Ecosystem overview, roles, and project lifecycles.
Days 3–4	**Mathematics for Data Science**	Linear algebra and calculus fundamentals.
Days 5–6	**Statistics – I**	Descriptive statistics, probability, and distributions.
Days 7–8	**Statistics – II**	Inferential statistics, hypothesis testing, and p-values.
Day 9	**Fundamentals of Computing**	Hardware/software basics and algorithmic thinking.
Days 10–11	**Introduction to Python**	Syntax, environment setup, and execution models.
Day 12	**Variables & Basic Types**	Variables, Data Types, and String manipulation.
Days 13–14	**Data Structures**	Mastery of Lists, Tuples, and Dictionaries.

Phase 2: Advanced Programming Concepts & Database Management

Duration: Weeks 3–4

Timeline	Topic	Key Focus Areas
Day 15	**Control Flow**	Conditional Statements (If-Else).
Day 16	**Iteration**	Looping structures (For, While).
Days 17–18	**Functional Programming**	Functions, arguments, and variable Scope.
Day 19	**Error Handling**	Modules and Exception Handling.
Day 20	**File I/O**	Working with external files in Python.
Days 21–22	**API Integration**	Using and consuming APIs within Python scripts.
Days 23–26	**Object-Oriented Programming**	OOP Fundamentals – Classes, Objects, and Inheritance.
Days 27–28	**SQL Basics**	Relational databases and basic CRUD operations.
Days 29–30	**Advanced SQL**	Complex Joins, aggregations, and subqueries.

Phase 3: Data Manipulation & Exploratory Analysis

Duration: Weeks 5–6

Timeline	Topic	Key Focus Areas
Days 31–34	**NumPy**	Numerical computing, arrays, and vectorization.
Days 35–40	**Pandas**	DataFrames, cleaning, merging, and transformation.
Days 41–42	**Basic Visualization**	Matplotlib and Seaborn (Static plotting).
Days 43–44	**Advanced Visualization**	Interactive charts and complex statistical plots.

Phase 4: Machine Learning Theory & Implementation

Duration: Weeks 7–8

Timeline	Topic	Format
Days 45–46	**Introduction to Machine Learning**	Supervised vs. Unsupervised Learning.
Days 47–50	**Linear Regression Theory**	4-Part Lecture series on mathematical foundations.
Days 51–52	**Linear Regression (Applied)**	Hands-on implementation using Scikit-learn.
Days 53–54	**Logistic Regression (Applied)**	Classification techniques using Scikit-learn.
Days 55–56	**Decision Trees**	Concept, splitting criteria, and implementation.
Days 57–58	**Random Forest**	Ensemble methods and bagging.
Days 59–60	**Gradient Boosting**	Boosting algorithms and performance tuning.
Days 61–62	**Unsupervised Learning**	Clustering (K-Means) and Recommender Systems.